

WASP-2b

R-0039

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_250918 · created 2026-07-08T23:24:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460936.75 +/- 0.0085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.136 +/- 0.046
Transit depth (Rp/Rs)^2	1.85 +/- 1.26 [%]
Orbital Inclination (inc)	84.86 +/- 2.89
Airmass coefficient 1 (a1)	1.67 +/- 0.69
Airmass coefficient 2 (a2)	-0.39 +/- 0.34
Scatter in the residuals of the lightcurve fit is	38.39 %
Best Comparison Star	#2 - [517, 117]
Optimal Method	PSF photometry
Transit Duration (day)	0.062 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.136 $\pm$ 0.046 vs 0.1326 (deviation 0.07 σ)
Duration fit vs published	0.062 d vs 0.0737 d (deviation 0.34 σ)
Mid-time O-C	20.6 min
Depth SNR	3.0
Residual scatter	38.39 %

Light curve

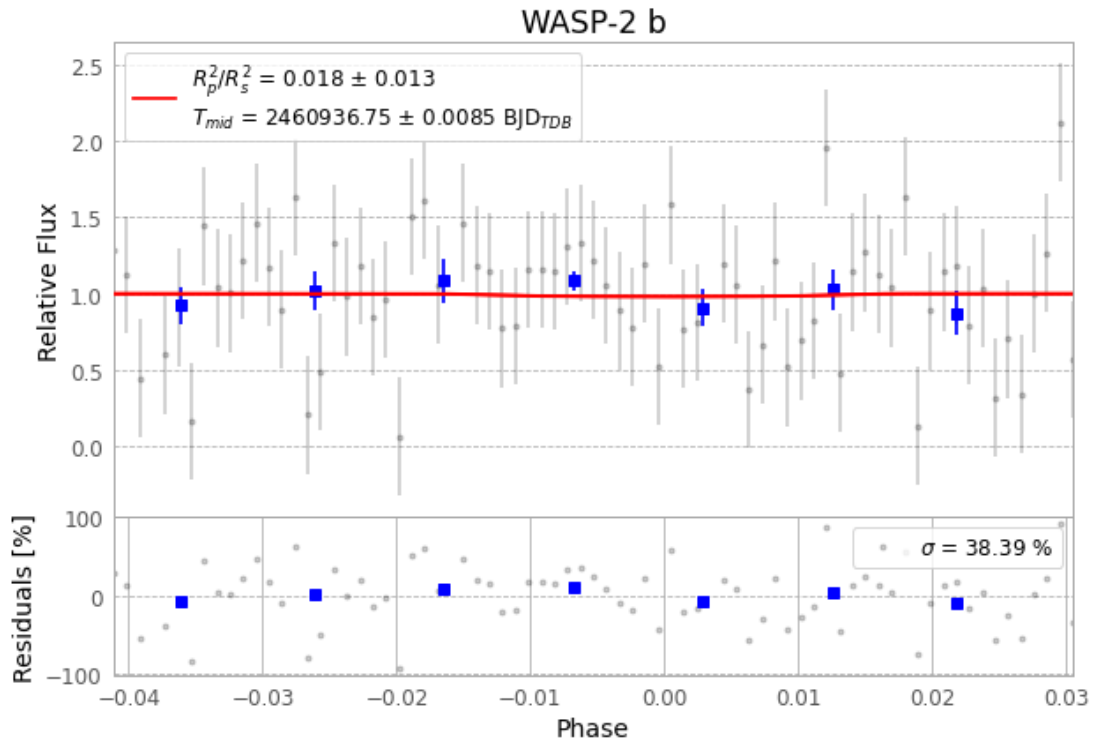


Figure 1 — FinalLightCurve_WASP-2 b_2025-09-17.png (SHA-256 2933f3bcd1b60e70...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [384, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e8b3fd9acde16efb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

75 FITS frames (49 MB), WASP-2250918034515.FITS → WASP-2250918072715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-250918.log (a3fd7a7163df...), FinalLightCurve_WASP-2 b_2025-09-17.png (2933f3bcd1b6...), FinalLightCurve_WASP-2 b_2025-09-17.csv (e32a6204f252...)

Compute resources

Wall time 162.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.